

CUDA Bitonic Sort Optimization

Kevin Bell

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1 Overview

This report summarizes timing results for a baseline bitonic sort implementation and three optimizations. The baseline source is `bitonic.sort.cu` and the modifications are `mod1.cu`, `mod2.cu`, and `mod3.cu`.

All timings measure kernel execution using the CUDA synchronization approach described in the assignment. The array size was 2^{30} values and all tests were compiled for a compute capability 8.6 device.

2 Results

The baseline kernel required 3501.20 ms. Table 1 lists the measured times and speedup relative to the baseline. Speedup is computed as $T_{\text{baseline}}/T_{\text{mod}}$.

Kernel	Time (ms)	Speedup
Baseline	3501.20	1.00
Modification 1	3497.79	1.00
Modification 2	1977.88	1.77
Modification 3	1749.70	2.00

Table 1: Kernel execution times for sorting 2^{30} values and speedup over the baseline.

Figure 1 visualizes the speedup achieved by each optimization.

3 Discussion

Modification 1 eliminates idle threads by having every CUDA thread operate on exactly one pair of indices.

Modification 2 merges multiple minor steps into a single kernel when possible, reducing kernel launch overhead.

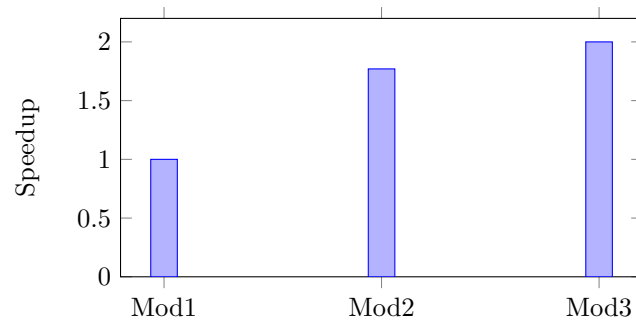


Figure 1: Speedup relative to the baseline implementation.

Modification 3 builds on the merged-kernel approach by keeping the j/set loop inside the kernel while each thread processes multiple index pairs to increase cache reuse. In these tests each thread handled four pairs.